21 Free Formula-Based Questions

This document contains 21 free questions allowing you to test your knowledge and understanding of the formulas needed for the Project Management Professional (PMP)® exam

Good luck!

Self-Assessment Questions

Question 1: If SV = -800 and PV = 7000, how much is EV?

A.) 6000
B.) 8000
C.) 6200
D.) 9000

Question 2: Which of the following are the most commonly used performance measures for evaluating whether or not work is being accomplished as planned at any given point of time?

A.) PV & AC
B.) CV & SV
C.) SPI, EV & AC
D.) AC,EV & PV

Question 3: If CPI is 1.03 and AC is $6,000, how much is EV?

A.) 200
B.) 7000
C.) 6180
D.) 7200

Question 4: There are 20 stakeholders on a project. What is the total number of potential communication channels in this project?

A.) 10
B.) 45
C.) 190
D.) 100

Question 5: You are managing a bridge construction project. Half of the project work has been completed. The total planned cost at this stage was $2.5 million. The actual physical work that has been completed at this stage is worth $2.4 million. You have already spent $5.0 million on the project. What is the Schedule Variance?

A.) $100,000
B.) -$100,000
C.) $2.6 million
D.) $2.5 million

Question 6: A task has a pessimistic estimate of 30 days, most likely estimate of 16 days, and optimistic estimate of 10 days. How much is the activity duration based on the PERT technique?

A.) 30 days
B.) 20 Days
C.) 18 Days
D.) 10 Days

Question 7: Your project is in an early stage and you need to estimate the overall duration. Based on your experience with similar past projects, you estimate a 6-month completion time. What is this type of estimation technique called?

A.) Three point estimating
B.) Bottom-up-estimating
C.) WAG
D.) Analogous

Question 8: You are managing a bridge construction project. You have completed half of the project work. The total planned cost at this stage is $500. The actual physical work that has been completed at this stage is worth $400. You have already spent $1,000 on the project. What is the CPI?

A.) 0.5
B.) 4
C.) 0.4
D.) 1

Question 9: The project management team is determining the activity dependencies to define the logical relationship between them. The project manager suggests that the team applies 10 days lag on a finish-to-start relationship. What does he actually mean?

A.) The successor activity can start ten days before the predecessor activity is completed.
B.) The successor activity cannot start until ten days after the predecessor activity is completed.
C.) The successor activity can only start ten days after the predecessor activity has started.
D.) The successor activity can only finish ten days after the predecessor activity has finished.

Question 10: A project manager finds that there is some variance in the project. What should the project manager do first?

A.) Communicate the variance to the project team
B.) Cancel the project
C.) Continue working and ignore the variance
D.) Identify the root cause of the variance

Question 11: A project started on April 1st, 2013 is expected to be completed by March 31st, 2014. During a project review on September 30th, 2013 it is determined that the total Actual Cost is $17.6 Million, the total Earned Value is $12.3 Million, and the total Planned Value is $12.8 Million. How much is the Schedule Variance?

A.) $5.2 Million
B.) $0.5 Million
C.) -$0.5 Million
D.) -$4.8 Million

Question 12: A project manager is required to anticipate the total final cost for a schedule activity. The past performance shows that the original estimating assumptions are no longer relevant to the present situation. Which of the following approaches should the project manager use to calculate the cost?

A.) EAC using new estimate
B.) EAC using remaining budget
C.) ETC based on new estimate
D.) ETC based on typical variances

Question 13: EV = 400. PV = 600. AC = 395. Which of the following is correct?

A.) CPI = 400 / 395
B.) SV = 400 * 600
C.) EAC = (395 * 400) - 600
D.) SPI = 600 / 400

Question 14: Mary is managing a data migration project. The project is in execution. The project's current total Earned Value (EV) is $25,000 and the current Schedule Variance (SV) is $5,000. What is the project's current Schedule Performance Index (SPI)?

A.) 1
B.) 1.25
C.) 0.83
D.) 0.2

Question 15: You are required to give your project sponsor a rough order of magnitude (ROM) estimate of the project costs. Based on your previous experience, you believe that the project will cost around $100,000. What should you present?

A.) $50,000 to $200,000
B.) $75,000 to $175,000
C.) $95,000 to $110,000
D.) $100,000

Question 16: A project's CPI is 0.83 and SPI is 1.25. Which of the following statements about this project must be true?

A.) EV is greater than AC
B.) EV is less than PV
C.) PV is less than AC
D.) PV is greater than AC

Question 17: A project's budget is $110,000. The total actual cost is $60,000 and the total earned value is $50,000. The total planned value of the project at this stage is $40,000. What is the project's new EAC if the project's cost performance is expected to continue at the current rate?

A.) $88,000
B.) $91,667
C.) $132,000
D.) $137,500

Question 18: An investor is considering building a new manufacturing factory. The total investment required to build the factory is $120 million. There is a 60% chance that the investor will get $200 million revenue out of the investment. However, due to uncertainty, there is a 40% chance that the investor will only get $90 million revenue. What is the expected value of the profit associated with this investment?

A.) $156 million
B.) $84 million
C.) $48 million
D.) $36 million

Question 19: You are managing a telecommunications project, and 10 telecom engineers report to you. Due to a change in scope requested by your customer, you need to add more team members, and you decide to hire two more engineers. How many communication channels will be added by hiring these engineers?

A.) 21
B.) 23
C.) 66
D.) 78

Question 20: You are preparing business cases for two telecommunications projects that your company is considering. The first of these is a $50 Million GSM technology-based project, while the second one is a $45 Million CDMA technology-based project. Both projects could be very lucrative and rewarding. However, your financial controller has told you that your company can only invest in one of these projects. If you choose to execute the GSM project, what will be your opportunity cost?

A.) $95 Million
B.) $50 Million
C.) $45 Million
D.) $5 Million

Question 21: You are managing a construction project. You have just finished defining your activities and you are now estimating their durations. Obtaining government permits is one of the project activities that is critical to the success of your project. You obtain expert opinion regarding time requirements for this activity, and you have determined that the duration will most likely be 15 days. It would be very optimistic to expect to obtain the permits in 10 days. In addition, taking the current environmental factors into consideration, it may take up to 30 days to obtain the permits. What estimate should you use for this activity if you want to put more weight on the most likely duration with a further 10 percent contingency reserve? (Round to the nearest day.)

A.) 17 Days
B.) 18 Days
C.) 20 Days
D.) 33 Days

Answers & Explanations

Question 1: If SV = -800 and PV = 7000, how much is EV?

A.) 6000
B.) 8000
C.) 6200
D.) 9000

Correct answer is **C**

Explanation: SV = EV - PV, therefore EV = SV + PV. So EV = -800 + 7000 = 6200.

Reference: PMBOK 5th Edition, Page 218

Question 2: Which of the following are the most commonly used performance measures for evaluating whether or not work is being accomplished as planned at any given point of time?

A.) PV & AC
B.) CV & SV
C.) SPI, EV & AC
D.) AC,EV & PV

Correct answer is **B**

Explanation: Performance is usually measured in terms of variance from planned values. CV (Cost Variance) and SV (Schedule Variance) are two most commonly used performance indicators.

Reference: PMBOK 5th Edition, Page 218

Question 3: If CPI is 1.03 and AC is $6,000, how much is EV?

A.) 200
B.) 7000
C.) 6180
D.) 7200

Correct answer is **C**

*Explanation: CPI = EV/AC. Therefore EV = CPI*AC = 1.03 * $6,000 = $6,180.*

Reference: PMBOK 5th Edition, Page 219

Question 4: There are 20 stakeholders on a project. What is the total number of potential communication channels in this project?

A.) 10
B.) 45
C.) 190
D.) 100

Correct answer is **C**

Explanation: Total number of communication channels can be calculated using the formula n(n – 1)/2, where n = number of stakeholders.

Reference: PMBOK 5th Edition, Page 292

Question 5: You are managing a bridge construction project. Half of the project work has been completed. The total planned cost at this stage was $2.5 million. The actual physical work that has been completed at this stage is worth $2.4 million. You have already spent $5.0 million on the project. What is the Schedule Variance?

A.) $100,000
B.) -$100,000
C.) $2.6 million
D.) $2.5 million

Correct answer is **B**

Explanation: SV=EV-PV =$2.4 million - $2.5 million = -$100,000

Reference: PMBOK 5th Edition, Page 218

Question 6: A task has a pessimistic estimate of 30 days, most likely estimate of 16 days, and optimistic estimate of 10 days. How much is the activity duration based on the PERT technique?

A.) 30 days
B.) 20 Days
C.) 18 Days
D.) 10 Days

Correct answer is **C**

Explanation: The PERT estimation formula is [Optimistic + (4 x Most Likely) + Pessimistic] / 6 = [10 + (4 x 16) + 30] / 6 = 104/6 = 17.33 days, so the best answer is 18 days.

Reference: PMBOK 5th Edition, Page 171

Question 7: Your project is in an early stage and you need to estimate the overall duration. Based on your experience with similar past projects, you estimate a 6-month completion time. What is this type of estimation technique called?

A.) Three point estimating
B.) Bottom-up-estimating
C.) WAG
D.) Analogous

Correct answer is **D**

Explanation: Analogous duration estimating means using the actual duration of a previous, similar schedule activity as the basis for estimating the duration of a future schedule activity.

Reference: PMBOK 5th Edition, Page 169

Question 8: You are managing a bridge construction project. You have completed half of the project work. The total planned cost at this stage is $500. The actual physical work that has been completed at this stage is worth $400. You have already spent $1,000 on the project. What is the CPI?

A.) 0.5
B.) 4
C.) 0.4
D.) 1

Correct answer is **C**

Explanation: CPI = EV/AC. EV = $400, AC = $1,000, PV = $500 - Therefore CPI = 0.4.

Reference: PMBOK 5th Edition, Page 219

Question 9: The project management team is determining the activity dependencies to define the logical relationship between them. The project manager suggests that the team applies 10 days lag on a finish-to-start relationship. What does he actually mean?

A.) The successor activity can start ten days before the predecessor activity is completed.
B.) The successor activity cannot start until ten days after the predecessor activity is completed.
C.) The successor activity can only start ten days after the predecessor activity has started.

D.) The successor activity can only finish ten days after the predecessor activity has finished.

Correct answer is **B**

Explanation: The finish-to-start dependency means the successor activity cannot start until the predecessor activity is finished. A 10-day lag will further delay the predecessor activity by 10 days.

Reference: PMBOK 5th Edition, Page 156

Question 10: A project manager finds that there is some variance in the project. What should the project manager do first?

A.) Communicate the variance to the project team
B.) Cancel the project
C.) Continue working and ignore the variance
D.) Identify the root cause of the variance

Correct answer is **D**

Explanation: The project manager must investigate the variance. It is the responsibility of the project manager to identify the root cause first in order to help prevent this variance in the future on the project.

Reference: PMI Code of Ethics & Professional Conduct: Responsibility

Question 11: A project started on April 1st, 2013 is expected to be completed by March 31st, 2014. During a project review on September 30th, 2013 it is determined that the total Actual Cost is $17.6 Million, the total Earned Value is $12.3 Million, and the total Planned Value is $12.8 Million. How much is the Schedule Variance?

A.) $5.2 Million
B.) $0.5 Million
C.) -$0.5 Million
D.) -$4.8 Million

Correct answer is **C**

Explanation: Schedule Variance = Earned Value - Planned Value = $12.3 Million - $12.8 Million = -$0.5 Million.

Reference: PMBOK 5th Edition, Page 218

Question 12: A project manager is required to anticipate the total final cost for a schedule activity. The past performance shows that the original estimating assumptions are no longer relevant to the present situation. Which of the following approaches should the project manager use to calculate the cost?

A.) EAC using new estimate
B.) EAC using remaining budget
C.) ETC based on new estimate
D.) ETC based on typical variances

Correct answer is **A**

Explanation: The project manager should determine the EAC with a new estimate. This approach is used only when the original estimating assumptions are o longer relevant in the present scenario.

Reference: PMBOK 5th Edition, Page 220

Question 13: EV = 400. PV = 600. AC = 395. Which of the following is correct?

A.) CPI = 400 / 395
B.) SV = 400 * 600
C.) EAC = (395 * 400) - 600
D.) SPI = 600 / 400

Correct answer is **A**

Explanation: CPI = EV / AC = 400 / 395 is correct. The rest of the formulas are incorrect.

Reference: PMBOK 5th Edition, Page 224

Question 14: Mary is managing a data migration project. The project is in execution. The project's current total Earned Value (EV) is $25,000 and the current Schedule Variance (SV) is $5,000. What is the project's current Schedule Performance Index (SPI)?

A.) 1
B.) 1.25
C.) 0.83
D.) 0.2

Correct answer is **B**

The formula for finding the correct answer is SPI = EV/PV. However, the problem is that while we know that EV = $25,000 we do not know the value for PV.. This means we first have to determine the current PV in order to determine the current SPI.
To determine the PV we use the SV formula. Since SV = EV-PV, this means that PV = EV-SV. Plugging in the EV and SV values we get PV = $25,000 - $5,000 = $20,000.
Now we can determine the current SPI as EV/PV = $25,000 / $20,000 = 1.25.

Reference: PMBOK Guide 5th Edition, Pages 218, 219

Question 15: You are required to give your project sponsor a rough order of magnitude (ROM) estimate of the project costs. Based on your previous experience, you believe that the project will cost around $100,000. What should you present?

A.) $50,000 to $200,000
B.) $75,000 to $175,000
C.) $95,000 to $110,000
D.) $100,000

Correct answer is **B**

Explanation: ROM is in the range of -25% to +75%. The PM must present the project costs as a ROM estimate from $75,000 to $175,000.

Reference: PMBOK 5th Ed, Page 201

Question 16: A project's CPI is 0.83 and SPI is 1.25. Which of the following statements about this project must be true?

A.) EV is greater than AC
B.) EV is less than PV
C.) PV is less than AC
D.) PV is greater than AC

Correct answer is **C**

Explanation: As CPI is less than 1, this means that the EV is less than AC. Similarly, since SPI is greater than 1, this means that EV is greater than PV. Since EV < AC and EV > PV, this implies that PV < EV < AC. Therefore PV < AC.

Reference: PMBOK 5th Ed, Page 219

Question 17: A project's budget is $110,000. The total actual cost is $60,000 and the total earned value is $50,000. The total planned value of the project

at this stage is $40,000. What is the project's new EAC if the project's cost performance is expected to continue at the current rate?

A.) $88,000
B.) $91,667
C.) $132,000
D.) $137,500

Correct answer is **C**

Explanation: The BAC is $110,000, AC is $60,000 and EV is $50,000. Since the current cost performance is expected to continue, we will calculate EAC as BAC/CPI. CPI = EV/AC = 50,000/60,000 = 0.833. EAC = BAC/CPI = 110,000/0.833 = $132,000.

Reference: PMBOK 5th Ed, Page 220

Question 18: An investor is considering building a new manufacturing factory. The total investment required to build the factory is $120 million. There is a 60% chance that the investor will get $200 million revenue out of the investment. However, due to uncertainty, there is a 40% chance that the investor will only get $90 million revenue. What is the expected value of the profit associated with this investment?

A.) $156 million
B.) $84 million
C.) $48 million
D.) $36 million

Correct answer is **D**

*Explanation: The EMV for the profit is calculated by multiplying the value of each possible profit outcome by its probability of occurrence and adding the products together. Hence the EMV = [($200m - $120m) * 60%] + [($90m - $120m) * 40%] = [$80m * 60%] + [-$30m * 40%] = $48m - $12m = $36m.*

Reference: PMBOK 5th Ed, Page 339

Question 19: You are managing a telecommunications project, and 10 telecom engineers report to you. Due to a change in scope requested by your customer, you need to add more team members, and you decide to hire two more engineers. How many communication channels will be added by hiring these engineers?

A.) 21
B.) 23

C.) 66
D.) 78

Correct answer is **B**

*Explanation: As 10 engineers reported to you before the recruitment exercise, the total project team size was 11 and the number of communication channels was 11*10/2 = 55. After the recruitment of two engineers the new team size is 13 and the communications channels are now 13*12/2 = 78. The additional communication channels therefore total 78-55 = 23.*

Question 20: You are preparing business cases for two telecommunications projects that your company is considering. The first of these is a $50 Million GSM technology-based project, while the second one is a $45 Million CDMA technology-based project. Both projects could be very lucrative and rewarding. However, your financial controller has told you that your company can only invest in one of these projects. If you choose to execute the GSM project, what will be your opportunity cost?

A.) $95 Million
B.) $50 Million
C.) $45 Million
D.) $5 Million

Correct answer is **C**

Explanation: Opportunity cost is the cost of a commercial decision and is regarded as the value of the alternative that is forgone. Since you have decided to go with the $50 Million GSM project and forego the $45 Million CDMA project, the opportunity cost of this decision is $45 Million.

Reference: Head First PMP 2nd Edition, page 330

Question 21: You are managing a construction project. You have just finished defining your activities and you are now estimating their durations. Obtaining government permits is one of the project activities that is critical to the success of your project. You obtain expert opinion regarding time requirements for this activity, and you have determined that the duration will most likely be 15 days. It would be very optimistic to expect to obtain the permits in 10 days. In addition, taking the current environmental factors into consideration, it may take up to 30 days to obtain the permits. What estimate should you use for this activity if you want to put more weight on the most likely duration with a further 10 percent contingency reserve? (Round to the nearest day.)

A.) 17 Days

B.) 18 Days
C.) 20 Days
D.) 33 Days

Correct answer is **B**

*Explanation: As you want to put more weight on the most likely estimate, you should use the PERT formula for your initial estimate rather than taking a simple average of the three values. Your PERT estimate will be (10 + 4*15 + 30) / 6 = 16.67 days. Adding a 10 percent contingency reserve to it will give 16.67*1.1 = 18.33 days. Rounding your estimate to the nearest day will give 18 days.*

